PLO2, C4, CLO # 3: Analyze system requirements and come up with high level conceptual data model.

Q1: [20 marks]

a. The management of GiveLife blood bank keeps donor and their donations info on paper. For each donor there is a donor card with personal information and information about donations. Following is an Example of a donor card. **Examine** this donor card and **lay out** an ER diagram to keep this information in DBMS.

Donor Card		
CPR	060275-1133	
Name	Rasmus Berg	
Address	hilltop 1	
Bloodtype	A pos.	
Donations		
Date	Collector	Controller
1/12/2012	ABG	SM
2/13/2012	SM	ADD
6/30/2013	EG	SM

b. **Analyze** the following system requirements for an airport management system and **lay out** an EER diagram:

- Every airplane has a registration number and each airplane is of a specific model.
- The airport accommodates a number of airplane models, and each model is identified by a model number (e.g., A-320) and has a capacity and a weight.
- A number of technicians work at the airport. You need to store the name, NIC, address, phone number, and salary of each technician.
- Each technician is an expert on one or more plane model(s), and his or her expertise may overlap with that of other technicians. This information about technicians must also be recorded.

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- Traffic controllers must have an annual medical examination. For each traffic controller, you must store the date of the most recent exam.
- All airport employees (including technicians) belong to a union. You must store the union membership number of each employee. You can assume that each employee is uniquely identified by a NIC.
- The airport has a number of tests that are used periodically to ensure that airplanes are still airworthy. Each test has a CAA test number, a name, and a maximum possible score.
- The CAA requires the airport to keep track of each time a given airplane is tested by a given technician using a given test. For each testing event, the information needed is the date, the number of hours the technician spent doing the test, and the score the airplane received on the test.

PLO4, C3, CLO # 4: Apply the rules that are needed to convert an ER/EER model into a corresponding relational data model

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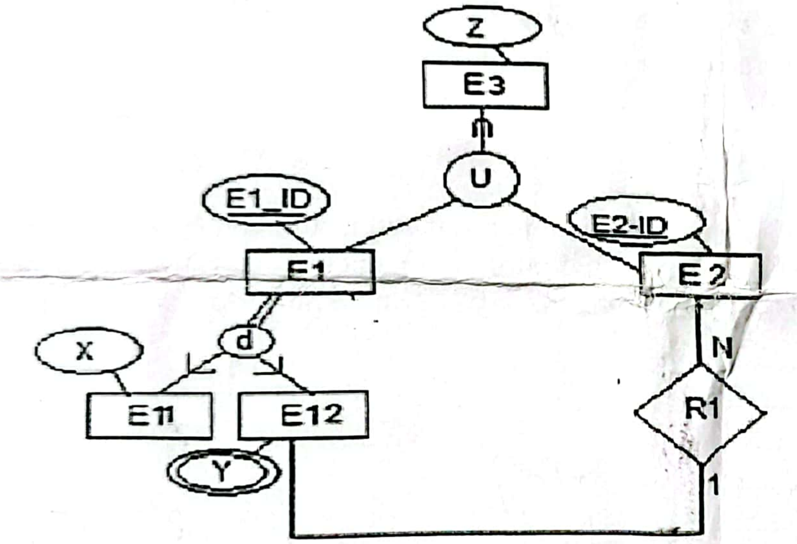
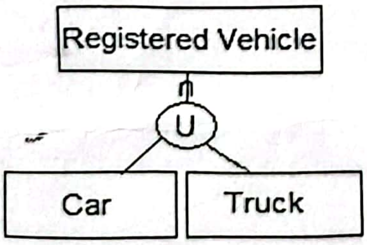
Q2: [10 marks]

a) **Derive** the relational model against the following EER diagram.

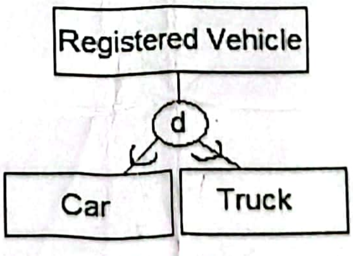
·) Union
·) Conversion
·) Weak entity type
) [redacted]



b) **Ascertain** if the following models, A and B equivalent to each other? Give proper reasoning for your answer. Assume all entities have required attributes.



Model A



Model B

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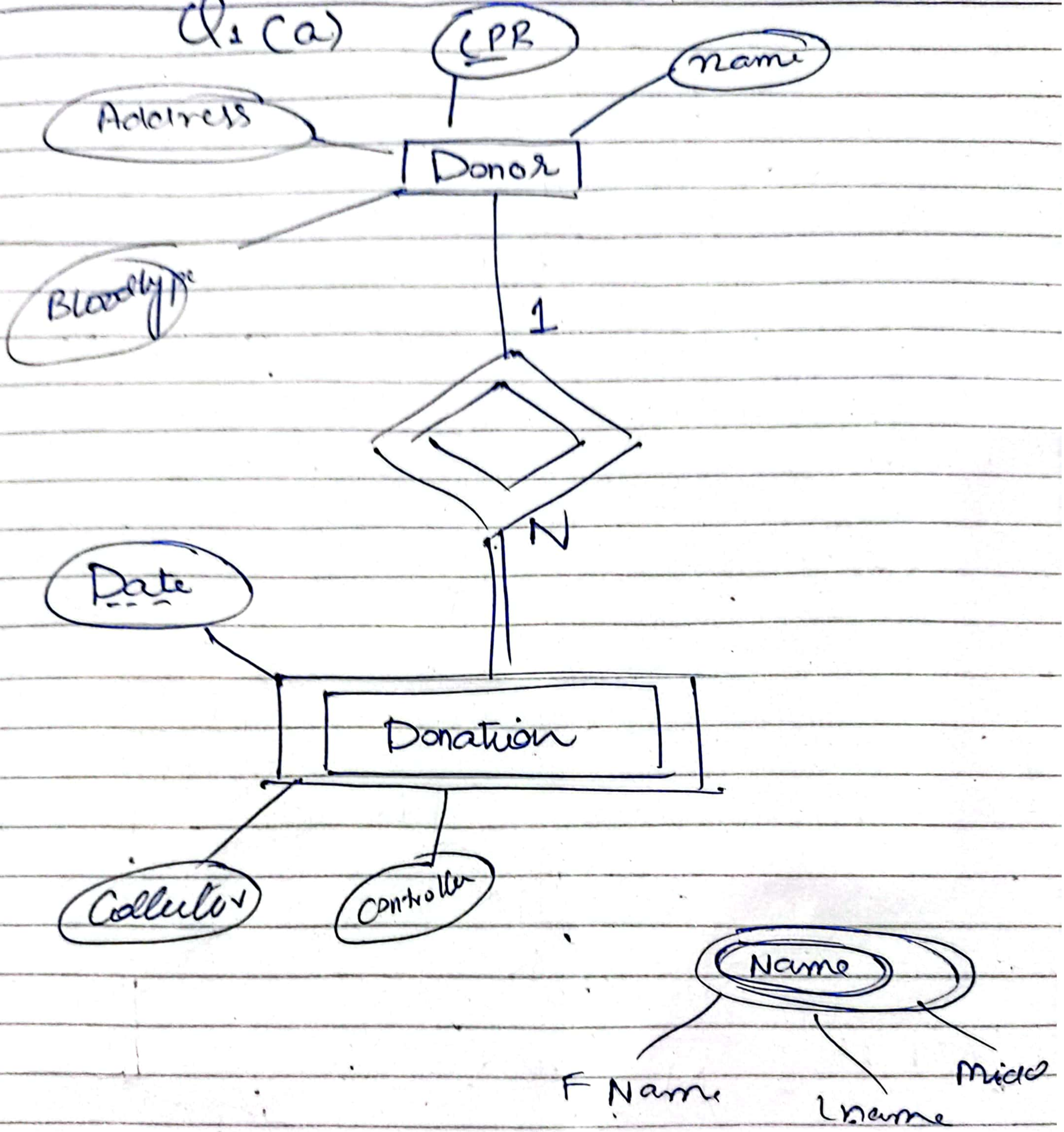
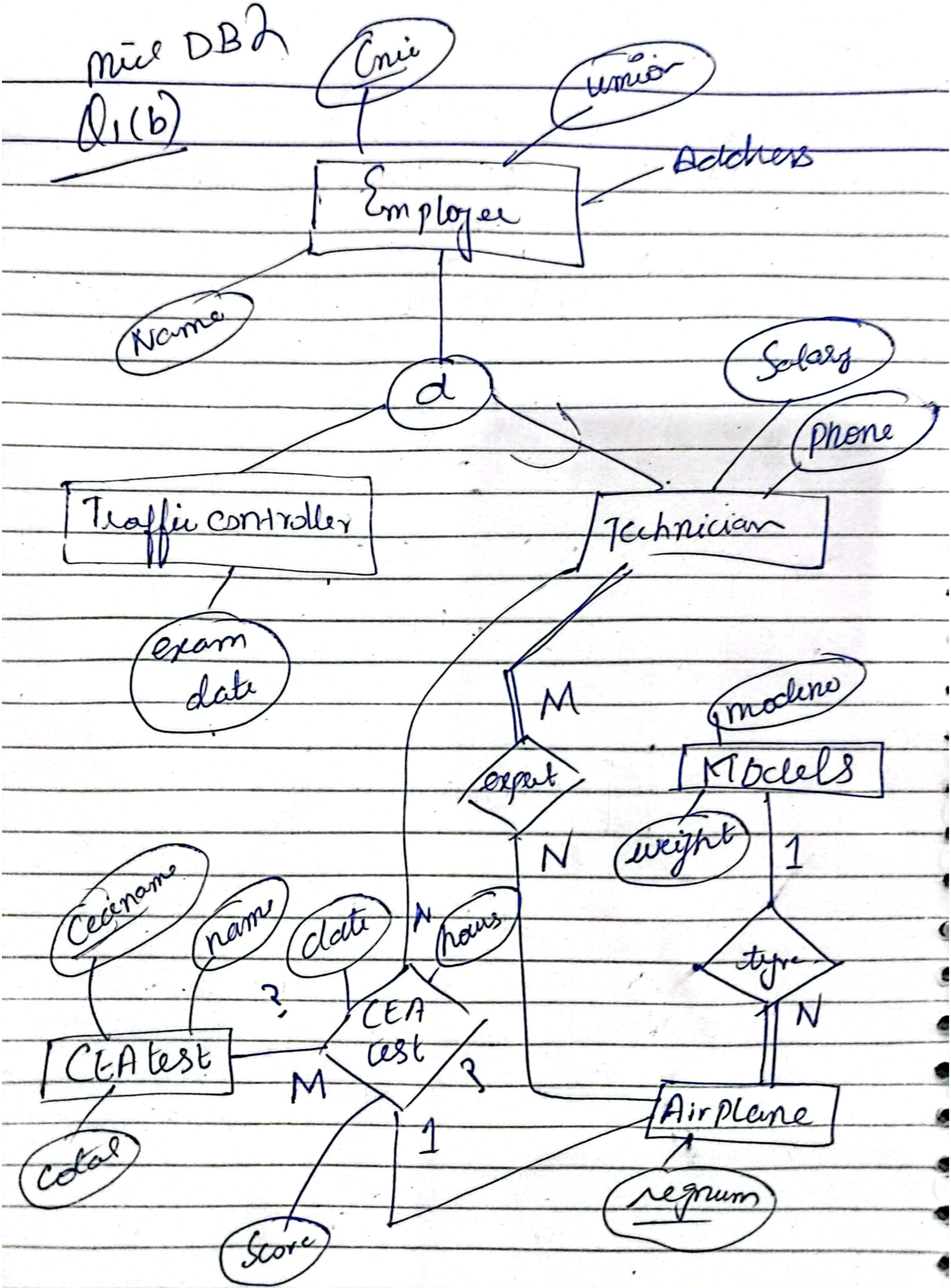
Mid2DB 2

Q1 (a)


CPR
name
Address
Donor
BloodType
1
N
Date
Donation
Collector
controller
Name
F Name
Lname
Midd

mid DB2

Q1(b)


Cnic
union
Address
Employee
Name
d
Salary
phone
Traffic controller
Technician
exam date
M
expert
N
modelno
Models
weight
1
type
N
Ceaname
name
date
N
hours
?
CEAtest
CEA test
M
?
1
Airplane
total
Score
regnum

Q2(a) DBMid

Option - A